An Investigation of SMOTE Based Methods for Imbalanced Datasets With Data Complexity Analysis (Extended Abstract)

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***Abstract*—This extended abstract highlights challenges with imbalanced datasets in real-world applications, where issues like noise, class overlap, and small subsets of data impact classification accuracy. While the Synthetic Minority Oversampling Technique (SMOTE) addresses imbalanced datasets by increasing minority class examples, it struggles with handling these data complexities and might worsen the situation. As a result, several SMOTE variants have emerged, aiming to improve its effectiveness by integrating it with other methods or altering its approach. This paper offers a comparative analysis of these variants, examining how each tackles specific data complexities. Through experiments on 24 imbalanced datasets, changes in complexity measures resulting from these SMOTE variants, in terms of F1-Score and data complexity metrics are observed and demonstrated.**

***Index Terms*—SMOTE, imbalanced learning, data complexity, imbalanced data, oversampling, classification**

I. Introduction

The evolution of computing infrastructure has transformed machine learning research from primarily theoretical models in the late 1990s [1] to a rapid prototyping-based investigations [2]. Regardless of the advancement in machine learning research, there are assumptions that need to be adhered (or at least used as guidelines) for machine learning methods to work effectively (which are built into many machine learning algorithms). Such assumptions are defined as follows:

Assumption 1. *The objective is focused on maximizing the machine learning model accuracy.*

Assumption 2. *Future data has similar distribution characteristics as training data, from which the machine learning model is derived from [3].*

Machine learning depends on these assumptions to address real-world issues, but these assumptions often face violations caused by uncertainties that arise during the learning process in real-world scenarios. To illustrate, many classic machine learning classifiers are based on the minimization of training error, such as:

$$J = \frac{1}{2n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 \tag{1}$$

where $\hat{y}_i$ is the predicted value and y_i is the true value. Based on Equation 1, if the data consists of 10 positive samples and 90 negative samples, a naive classifier will just predict everything to be negative, so that it can achieve 90% classification accuracy, instead of predicting otherwise, which will only result in 10% accuracy.

Basically, class imbalance problem can be solved by two basic strategies. The first strategy is based on data-level approach, that is, through undersampling majority class or oversampling minority class [4]. The second approach is based on algorithmic approach, that is through various model selection or optimization tasks [5].

From statistical and computation standpoints, data-level approach such as undersampling, oversampling, or combination of both are much more flexible, faster to compute and much more verbose, compared to algorithmic-level approach. Hence, in this paper, our focus will be on data-level approach, in particular, on a technique called as **S**ynthetic **M**inority **O**versampling **TE**chnique (**SMOTE**) [6].

II. Background

SMOTE [6] is the most well-known oversampling technique that have been proposed to even the data distribution between majority and minority classes. The technique has been successfully used to solve the imbalanced dataset problem in various researches and real-world domains [7]. From its first emergence in 2002 till now, SMOTE technique has been cited by more than 27,000 scholarly manuscripts[1].

In addition to that, many have performed extensive study on SMOTE alone. Such studies include performance aspects

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[1] According to Google Scholar.

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of SMOTE, time complexity, application of different operating principles and challenges that emerge [7].

Our work also can be considered as extensive study on SMOTE based techniques. However, unlike previous studies, we expand the analysis of SMOTE performance by also including data complexity metrics into our investigation, instead of merely observing the performance evaluation metrics.

III. Findings

A. Objective

Regardless of the severity of the imbalance degree in particular dataset, most modern machine learning classifiers are capable to distinguish minority patterns from majority patterns, provided the dataset is in a perfect shape. If the dataset suffers from noises, small disjuncts or class overlapping, the difficulty of the learning process could be increased, leading to a sub-optimal prediction model. This is why we include data complexity as part of our study, so that we could investigate:

1) The capability of SMOTE variants in reducing the data complexity of imbalance dataset. Two data complexity metrics are used to evaluate this objective, which are Fisher's ratio and N1.
2) The classification performance in regards to data complexity metric, based on before and after SMOTE processing is being performed on imbalance dataset.

B. Experimentation Design

We selected 24 binary class imbalanced datasets as our datasets, with variety of instances and imbalance ratio value. Based on these datasets, we evaluated the capability of 8 selected SMOTE variants in reducing data complexity as well as the classification performance that were gained after SMOTE processing.

C. Analysis of Results

We evaluated the conducted experiments based on: 1) the change in F1-Score (prior/baseline and after SMOTE processing) and 2) the change in data complexity measure in relation the change in F1-Score (prior/baseline and after SMOTE processing). These two findings are illustrated in Fig. 1 and Fig. 2.

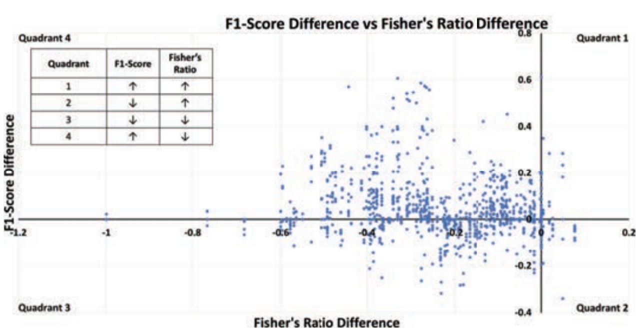

Fig. 1: The figure shows the Fisher's ratio and F1-Score changes from baseline to classification with SMOTE processing. Each quadrant illustrates the magnitude of change for each metric.

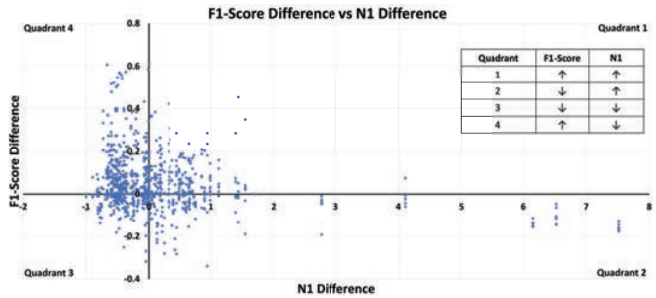

Fig. 2: The figure shows the N1 and F1-Score changes from baseline to classification with SMOTE processing. Each quadrant illustrates the magnitude of change for each metric.

We also computed the correlation between changes in Fisher's Ratio and N1 complexity metrics towards changes in F1-Score, such as in Table I.

TABLE I: Correlation analysis for changes in Fisher's Ratio and N1 complexity metrics towards changes in F1-Score.

Complexity Metric	Correlation to F1-Score
Fisher's Ratio	-0.15529
N1	-0.20685

The experimentation results can be summarized at least into two main conclusions, such as follows:

1) SMOTE can increase the classification performance of machine learning model on imbalanced datasets, **AND** also reducing the complexity of those datasets.
2) The negative correlation shown in Table I showcased that as data complexity decreases, the classification performance (such as based on F1-Score) increases. In the case of Table I, reducing N1 data complexity metric would have positive impact on the classification performance.

Further analysis and discussion of the results can be found in our paper [8].

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